

Data Science Masters

Plotly — Assignment Answers

Q1. Load the titanic dataset using Seaborn and use Plotly Express to plot a scatter plot for age and fare.

The titanic dataset is loaded using Seaborn. Plotly Express is then used to create a scatter plot with age on the x-axis and fare on the y-axis.

Google Colab Code:

```
import seaborn as sns
import plotly.express as px

titanic = sns.load_dataset("titanic")

fig = px.scatter(
    titanic,
    x="age",
    y="fare",
    title="Titanic: Age vs Fare"
)

fig.show()
```

Q2. Using the tips dataset, plot a box plot using Plotly Express.

A box plot summarizes the distribution of a numerical variable and helps compare values across categories. The following code uses the tips dataset and Plotly Express to create a box plot of total_bill by day.

Google Colab Code:

```
import plotly.express as px

tips = px.data.tips()

fig = px.box(
    tips,
    x="day",
    y="total_bill",
    title="Total Bill by Day"
)

fig.show()
```

Q3. Using the tips dataset, plot a histogram for x='sex' and y='total_bill'. Use smoker with pattern_shape and day with color.

The histogram uses sex on the x-axis and total_bill on the y-axis. The smoker column is used with pattern_shape and day is used with color, as required.

Google Colab Code:

```
import plotly.express as px

tips = px.data.tips()

fig = px.histogram(
    tips,
    x="sex",
    y="total_bill",
    color="day",
    pattern_shape="smoker",
    title="Total Bill by Sex, Day and Smoker"
)
```

```
)  
  
fig.show()
```

Q4. Using the iris dataset, plot a scatter matrix with species as the color parameter.

The scatter matrix shows pairwise relationships between the four specified iris measurements. The species column is used as the color parameter to distinguish species.

Google Colab Code:

```
import plotly.express as px  
  
iris = px.data.iris()  
  
fig = px.scatter_matrix(  
    iris,  
    dimensions=[  
        "sepal_length",  
        "sepal_width",  
        "petal_length",  
        "petal_width"  
    ],  
    color="species",  
    title="Iris Scatter Matrix"  
)  
  
fig.show()
```

Q5. What is Distplot? Using Plotly Express, plot a distplot.

A distplot (distribution plot) is used to visualize the distribution of a numerical variable. It commonly combines information such as a histogram and a density curve.

Plotly Express does not provide a function literally named distplot. A distribution plot can be created with Plotly Express using a histogram and a density curve.

Google Colab Code:

```
import plotly.express as px  
  
tips = px.data.tips()  
  
fig = px.histogram(  
    tips,  
    x="total_bill",  
    marginal="rug",  
    histnorm="probability density",  
    title="Distribution Plot of Total Bill"  
)  
  
fig.show()
```

Note: The assignment asks for the work to be created in a Jupyter notebook, uploaded to GitHub, and shared through the dashboard with a public repository.